

AP Precalculus — Unit 3 Practice Exam

Lessons 3.1–3.6

Name: _____

Date: _____

Section I: Multiple Choice (NO CALCULATOR)

1. The height of a point on a dolphin swimming in the ocean is modeled by a sinusoidal function $h(t)$. The dolphin reaches a maximum height of 3 feet above the surface and a minimum height of 3 feet below the surface. The time from one maximum height to the next is 4 seconds.

Which of the following could represent $h(t)$?

- (A) $3 \sin\left(\frac{\pi}{2}t\right)$
- (B) $3 \sin(\pi t)$
- (C) $\sin\left(\frac{\pi}{2}t\right) + 3$
- (D) $\sin(\pi t) + 3$

2. What is the amplitude of the function

$$f(x) = -4 \cos x + 1?$$

- (A) -4
- (B) 1
- (C) 3
- (D) 4

3. What is the period of

$$f(x) = \sin(3x)?$$

- (A) 2π
- (B) π
- (C) $\frac{2\pi}{3}$
- (D) $\frac{\pi}{3}$

4. Consider a circle centered at the origin. An angle of measure $\frac{4}{5}$ radians subtends an arc of length 40 units. What is the radius of the circle?

- (A) 5π
- (B) 10π
- (C) 50
- (D) 80

5. What is the midline of the function

$$f(x) = 2 \sin x - 5?$$

- (A) $y = 2$
- (B) $y = -5$
- (C) $y = -3$
- (D) $y = 5$

6. Which transformation maps $y = \sin x$ to $y = \sin(2x)$?

- (A) Vertical stretch by factor 2
- (B) Horizontal stretch by factor 2
- (C) Horizontal compression by factor 2
- (D) Vertical compression by factor 2

7. Let $f(x) = \sin x$. The function g has the same amplitude as f and a period that is one-half the period of f .

Which of the following defines g in terms of f ?

- (A) $g(x) = f\left(\frac{1}{2}x\right)$
- (B) $g(x) = \frac{1}{2}f(x)$
- (C) $g(x) = f(2x)$
- (D) $g(x) = 2f(x)$

8. What is the maximum value of

$$f(x) = 5 \sin x - 2?$$

- (A) 3
- (B) 5
- (C) 7
- (D) -2

9. What is the range of

$$f(x) = -3 \cos x + 1?$$

- (A) $[-3, 3]$
- (B) $[-2, 4]$
- (C) $[-4, 2]$
- (D) $[-1, 5]$

10. Which equation represents a phase shift of π units to the right of $y = \cos x$?
- (A) $y = \cos(x + \pi)$
(B) $y = \cos(x - \pi)$
(C) $y = -\cos x$
(D) $y = \cos(\pi x)$
11. Which situation is best modeled by a sinusoidal function?
- (A) The height of a Ferris wheel rider over time
(B) The distance a car travels at constant speed
(C) The population of bacteria doubling hourly
(D) The value of a savings account earning compound interest

Section II: Free Response Questions

Question 1 (NO CALCULATOR)

A function is defined by

$$f(x) = 3 \sin(2x) - 4.$$

- A. State the amplitude, period, and midline of f .
- B. Sketch one full cycle of $f(x)$.
- C. Determine the maximum, minimum, and points of inflection of $f(x)$.

Question 2 (CALCULATOR)

A Ferris wheel has a radius of 15 meters and its center is 18 meters above the ground. The wheel completes one full rotation every 30 seconds.

- (A) Write a sinusoidal function that models the height of a rider as a function of time.
- (B) Find the maximum and minimum heights of the rider.
- (C) Determine the times during the first rotation when the rider is 18 meters above the ground.
- (D) Explain why a sinusoidal model is appropriate for this situation.